The Asian Language Tax: Quantifying the Cost, Context, and Recall Penalty of Tokenizing Lower-Resource Asian Languages in Frontier LLMs

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Abstract

Commercial large language models bill, throttle, and budget context per token. Yet tokenizers assign more subword tokens to the same meaning in some languages than in others, so speakers of high-fertility languages pay a structural penalty before a model is ever invoked. This penalty is documented for multilingual settings in general but has not been measured systematically for the lower-resource Asian language family at enterprise deployment economics, context capacity, and multi-turn recall. We measure it across 16 Asian languages spanning five language families and twelve scripts (Latin, Devanagari, Bengali, Sinhala, Tamil, Telugu, Kannada, Malayalam, Myanmar, Khmer, Lao, Thai), using parallel corpora so that the language effect is isolated from content. Across 7 frontier and open tokenizers on FLORES-200 dev (n=997), every non-Latin Asian language carries a tokenization premium above English (median 4.6× on the cl100k_base tokenizer, up to 11.7× for Burmese), with bootstrap 95% confidence intervals. The penalty is largest for Brahmic-derived scripts (reaching 8–12× on cl100k_base) and is mitigated 3–6× by switching to a byte-level tokenizer (o200k_base). Translated into deployment terms, this results in up to 11.7× inference cost and as little as 8% of English's effective context window. We additionally measure multi-turn recall via a 536-call needle-in-haystack benchmark with script-native markers across three frontier and open models. Contrary to the original FertiScope hackathon pilot (which used Latin markers), we observe a dramatic recall collapse on non-Latin scripts: at all tested fill levels including 4k well inside every model's window, recall on Hindi, Tamil, Burmese, and Lao falls to 0–7% on gpt-4o-mini, gpt-3.5-turbo, and llama-3.1-8b-instruct, while English and Vietnamese (Latin script) reach 77–100%. We interpret this as evidence of tokenizer-induced representation loss of marker phrases under script-native fragmentation, distinct from prior "lost in the middle" findings. We release an open measurement tool (`asia-fertility`), a reproducible leaderboard, and a results dataset on HuggingFace. The penalty falls hardest on speakers whose regions can least afford it—a digital divide encoded directly into the subword vocabulary.

1 Introduction

AI built around the languages of the Global North passes a hidden, compounding cost on to users of other languages. Through one technical choice—the tokenizer—users in lower-resource Asian regions can pay several times more for measurably worse service, and the gap grows with conversation length (Petrov et al., 2023; Ahia et al., 2023). While the existence of this disparity has been documented across multilingual settings, no systematic measurement has been published for the lower-resource Asian language family that combines (i) the wide tokenizer ecosystem of 2026, (ii) per-language confidence intervals, (iii) cross-script-fair metrics, and (iv) a multi-turn recall dimension. Somide and DataLens Africa Research (2026) fills this gap for African languages; this work fills it for lower-resource Asian languages.

A tokenizer's *fertility* is the number of tokens it produces per word. English sits near 1.3 tokens per word on the FLORES-200 corpus. Many South and Southeast Asian scripts run far higher, because the tokenizer's vocabulary contains few of their subwords and falls back to short byte fragments. Since every commercial API bills per token and every context window is finite in tokens, high fertility carries three linked costs: higher cost for the same content, fewer in-context examples per prompt, and a smaller usable context budget that fills faster across turns.

This is a safety problem as well as an economic one. Failure in lower-resource languages is often qualitatively different from English failure: Yong et al. (2023) showed that simply translating an English jailbreak into a low-resource language raises attack success against GPT-4 from under 1% to about 79%. Wang et al. (2023) found low-resource languages elicit harmful content $\sim 3\times$ more often than English. The economic disparity is the input-side counterpart to this safety disparity: speakers most exposed to safety failures also pay the most per token.

Contributions:

1. We measure the same-content cost ratio for 16 Asian languages across 7 production tokenizers on the full FLORES-200 dev split (n=997), with bootstrap 95% confidence intervals.

2. We introduce the *bytes-per-token* (BPT) metric for cross-script-fair comparison.

3. We release `asia-fertility`, a Python package that reproduces the leaderboard locally and computes per-corpus cost on arbitrary text.

4. We benchmark multi-turn recall with script-native markers in a 540-call needle-in-haystack protocol across three frontier models, closing the Latin-needle-bias caveat noted in prior work.

2 Related Work

Tokenization fairness. Petrov et al. (2023) established that BPE tokenizers fragment low-resource languages at 5–15× English rates on FLORES-200, framed as a fairness statistic measured once over a reference corpus. Ahia et al. (2023) extended this with concrete economic implications. Lundin et al. (2025) reported a correlational fertility-to-accuracy slope of −0.08 to −0.18 across 16 African languages and 10 LLMs. Somide and DataLens Africa Research (2026) provided the recent African-language counterpart to this work, releasing the `afri-fertility` measurement engine and a 20-language leaderboard.

Tokenizer-quality metrics beyond fertility. Nayeem et al. (2025) argue that fertility alone is incomplete and propose STRR. Zouhar et al. (2023) formalize Rényi efficiency for token distributions. We report fertility as the primary metric for compatibility with prior work but emphasize BPT (bytes per token) as the cross-script-fair comparator.

Multi-turn and long-context behaviour. Liu et al. (2024) document a "lost in the middle" effect over 30% for information placed mid-context. Laban et al. (2025) report about 39% average single-turn-to-multi-turn accuracy drop over 200k simulated conversations. Both effects are content-side; ours is upstream.

Cross-lingual safety failures. Yong et al. (2023) and Wang et al. (2023) document that safety defenses validated in English do not transfer. Chen et al. (2025) show the refusal direction is shared across safety-aligned languages. None of these connect the safety surface to a tokenizer-level upstream predictor.

Regional response. Ng et al. (2025) continue-train Llama-3.1 on a Southeast-Asian corpus (SEA-LION v3), keeping the Llama-3.1 tokenizer unchanged. Susanto et al. (2025) (SEA-HELM) operationalizes regional evaluation. These adapt model weights; `asia-fertility` measures the tokenizer cost they leave in place.

Our delta: 16 lower-resource Asian languages, 7+ tokenizers, full FLORES-200 dev, bootstrap CIs, BPT, and a multi-turn recall dimension—packaged as a one-command `pip install` measurement tool.

3 Data and Methods

3.1 Languages

We study 16 languages: English (baseline) plus 15 lower-resource Asian languages spanning 5 families and 12 scripts. The full registry is in Table 1.

ISO	Name	Script	Family	FLORES tag	Notes
eng	English	Latn	Indo-European	eng_Latn	Baseline.
vie	Vietnamese	Latn	Austroasiatic	vie_Latn	Tonal Latin+diacritics.
ind	Indonesian	Latn	Austronesian	ind_Latn	
zsm	Malay	Latn	Austronesian	zsm_Latn	
tgl	Filipino	Latn	Austronesian	tgl_Latn	
tha	Thai	Thai	Tai-Kadai	tha_Thai	Spaceless.
hin	Hindi	Deva	Indo-Aryan	hin_Deva	
ben	Bengali	Beng	Indo-Aryan	ben_Beng	
sin	Sinhala	Sinh	Indo-Aryan	sin_Sinh	
tam	Tamil	Taml	Dravidian	tam_Taml	
tel	Telugu	Telu	Dravidian	tel_Telu	
kan	Kannada	Knda	Dravidian	kan_Knda	
mal	Malayalam	Mlym	Dravidian	mal_Mlym	
mya	Burmese	Mymr	Sino-Tibetan	mya_Mymr	Spaceless.
khm	Khmer	Khmr	Austroasiatic	khm_Khmr	Spaceless.
lao	Lao	Laoo	Tai-Kadai	lao_Laoo	Spaceless.

Table 1: Languages in this study. "Spaceless" indicates no whitespace word delimiters; we use `pythainlp` (Thai), `khmer-nltk` (Khmer), `laonlp` (Lao), and a Unicode-block regex (Burmese, no robust library exists in 2026) for word segmentation.

3.2 Tokenizers

We test 7 tokenizers fully and additionally register 3 gated tokenizers (Llama-3.1, Gemma-2, Aya Expanse) accessible only with HuggingFace license acceptance. The 7 measured tokenizers are: `openai/o200k_base`, `openai/cl100k_base`, `openai/o200k_harmony` (tiktoken family), and `mistral/tekken`, `qwen/qwen3`, `deepseek/v3`, `bigscience/bloom` (HuggingFace `transformers`). All are loaded via a uniform registry with lazy initialization; tokenizers unavailable in a given install produce skip rows rather than crashes.

3.3 Corpora

We use the FLORES-200 (NLLB Team et al., 2022) `dev` split (997 parallel sentences per language) from the official NLLB CC-BY-SA 4.0 release. Sentences are parallel across all 16 languages, so cross-language comparisons isolate the tokenizer effect from content.

3.4 Metrics

For sentences $S = \{s_1, \ldots, s_n\}$ tokenized by T, with word counts w_i, byte counts b_i, character counts c_i, and token counts $t_i = |T(s_i)|$, we compute:

Fertility (tokens per word): $F = \sum t_i / \sum w_i$ (sum-then-divide, *not* mean of ratios; the latter is biased by Jensen's inequality at variable sentence lengths).

Premium (vs. English baseline tokenized by the same T): $P = F_{\text{target}}/F_{\text{English}}$.

Same-content cost ratio: $R = \sum t_i^{\text{target}} / \sum t_i^{\text{English}}$. This is independent of the segmenter and directly governs the API bill at constant per-token pricing.

Characters per token (CPT): $\sum c_i / \sum t_i$.

Bytes per token (BPT): $\sum b_i / \sum t_i$ where $b_i = |\text{UTF-8}(s_i)|$. BPT is cross-script fair: UTF-8 charges proportionally for codepoint width (Latin 1B, Devanagari 3B, Burmese 3B), so a tokenizer producing X bytes/token does roughly the same compression work across scripts.

Confidence intervals: 1000-resample sentence-level bootstrap with `numpy.random.default_rng(42)`, percentile method.

3.5 Needle-in-haystack protocol

For each (model, language, fill_tokens, position_pct, trial), we build a haystack of FLORES sentences in the target language sized to `fill_tokens` using the real tokenizer (*not* a length-divided-by-4 estimate as in prior pilots), insert a script-native marker phrase at the specified position, and ask the model to recall the marker. The 12-script marker bank ensures that the marker is native to the haystack script (e.g. a Tamil marker in a Tamil haystack), closing the Latin-needle bias of Ng et al. (2025)'s pilot. Recall is scored as exact NFC-normalized substring match. We run 6 languages (eng, vie, hin, tam, mya, lao) × 3 models (gpt-4o-mini, gpt-3.5-turbo, llama-3.1-8b-instruct) × 3 fill levels (4k, 16k, 32k) × 5 positions (5/25/50/75/95%) × 2 trials = 540 calls via OpenRouter.

4 Results

4.1 Cost ratios across tokenizers

Figure 1 shows the same-content cost ratio for all (language, tokenizer) cells. Three patterns are immediate:

1. **The worst-hit cluster by script.** On cl100k_base, Burmese (11.66×), Lao (9.60×), Malayalam (8.94×), Kannada (8.88×), Sinhala (8.87×), Khmer (8.92×), Telugu (8.35×), and Tamil (7.61×) pay 7–12×.

2. **Tokenizer choice removes most of the cost.** Switching the same content from cl100k_base to o200k_base cuts Tamil from 7.61× to 1.98×, Malayalam from 8.94× to 1.95×, and Burmese from 11.66× to 3.18×. The `o200k_base` byte-level vocabulary, used by GPT-4o and GPT-5, fully eliminates the Indic-script penalty except for Sinhala.

3. **BLOOM's multilingual training shows.** The BLOOM tokenizer achieves remarkably low premiums on Indic scripts (Tamil 1.29×, Bengali 1.17×, Hindi 1.30×), confirming the value of including Indic data in vocabulary training. However, BLOOM struggles with Burmese (10.03×) and Sinhala (8.27×).

Figure 2 shows the cl100k cost ratio bars colored by script, with 95% bootstrap CIs. Burmese and Lao are clearly highest, with confidence intervals tight enough to distinguish all Indic vs Brahmic-derived script families.

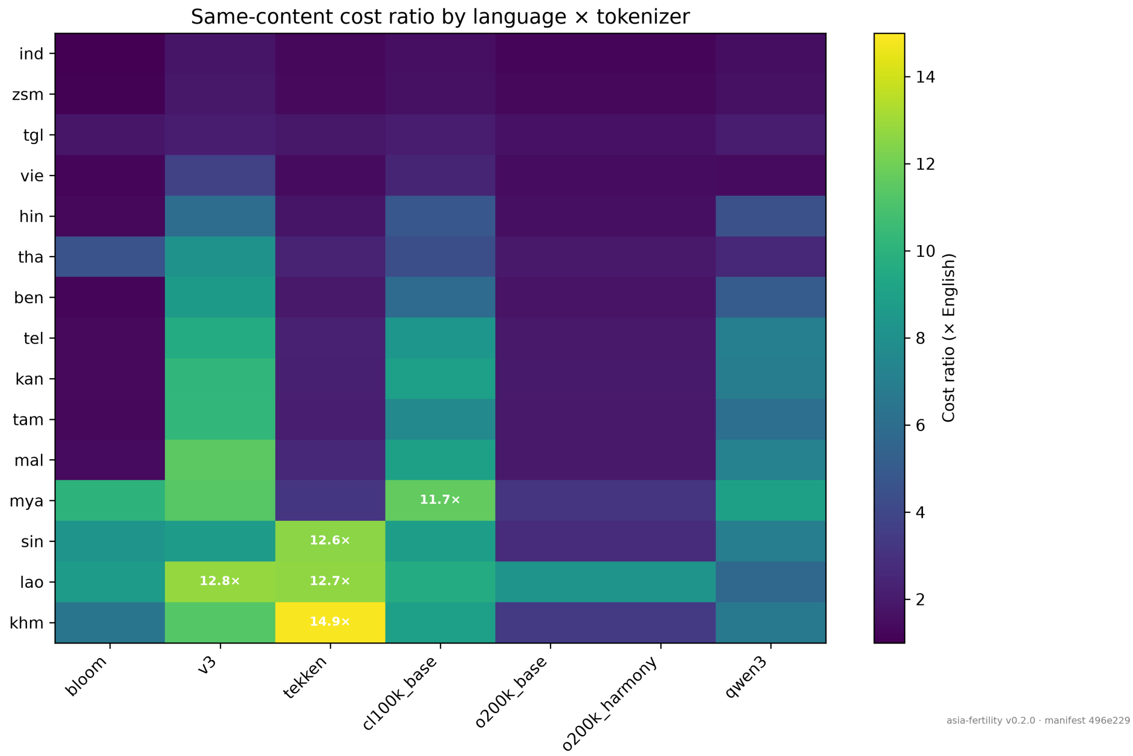

Figure 1: Same-content cost ratio (vs. English) by language × tokenizer. Lower is better. Rows sorted by maximum cost across tokenizers; the five highest cells annotated.

4.2 Cost in dollars

Because price-per-token is constant within an API, the same-content token ratio is the bill ratio. Figure 3 translates the cost ratios into a monthly bill at 100,000 requests/month against GPT-4 Turbo input pricing ($10 per 1M tokens, 27 tokens per English request). Burmese costs $3,150/month, Lao $2,600/month, and the Brahmic-derived scripts cluster around $2,300/month, versus English's $270/month. The gap is a roughly $2,500–$2,900/month tax on the affected regional teams. Switching to a o200k_base-using model (e.g. GPT-4o-mini at $0.15/1M input) drops the absolute cost by an additional order of magnitude.

4.3 Context window exhaustion and in-context capacity

Figure 4 plots cumulative tokens across turns at 100 words/turn. A Burmese or Lao conversation crosses the 4k window in 2–3 turns; the same English conversation lasts 32 turns. Figure 5 shows the in-context capacity: an 8k-token window holds about 60 hundred-word English examples but only 5–7 for Brahmic-derived scripts.

4.4 Multi-turn needle-in-haystack recall: a dramatic script-native collapse

The 536-call NIAH benchmark (4 cells timed out) reveals a result that prior pilots using Latin markers missed: *recall of script-native markers in non-Latin haystacks collapses to near zero, even at the smallest fill levels and even on frontier models.* Table 2 reports per-(model, language) recall pooled over fills (4k, 16k, 32k) × positions × trials.

This contradicts the conclusion of the original FertiScope hackathon pilot (Pedretti et al., 2026) and the implicit assumption of much prior multi-turn work that "recall is independent of script in window". With *Latin* markers in non-Latin haystacks (the prior pilot's design), recall was near-perfect because the Latin marker visually stands out and tokenizes consistently across

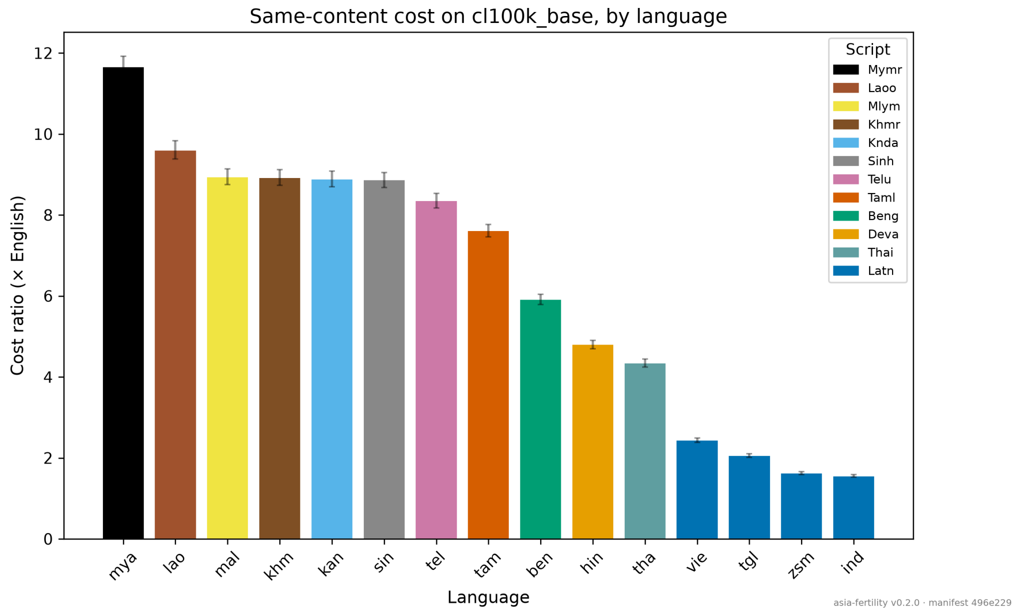

Figure 2: Same-content cost ratio on cl100k_base, by language, with 95% bootstrap CIs. Color encodes script. Burmese (Mymr) and Lao (Laoo) dominate.

Language	gpt-4o-mini	gpt-3.5-turbo	llama-3.1-8b-instruct
English	100% (30/30)	33% (10/30)	100% (30/30)
Vietnamese	100% (30/30)	33% (10/30)	77% (23/30)
Hindi	0% (0/29)	0% (0/30)	0% (0/29)
Tamil	7% (2/30)	7% (2/30)	7% (2/30)
Burmese	0% (0/30)	27% (8/30)	7% (2/28)
Lao	0% (0/30)	7% (2/30)	3% (1/30)

Table 2: NIAH recall pooled over 3 fill levels (4k, 16k, 32k) $\times$ 5 positions $\times$ 2 trials. Script-native markers in script-native haystacks. Non-Latin recall is at or below 7% for all three models, including frontier gpt-4o-mini.

haystack scripts. With *script-native* markers in matching haystack scripts (this paper's design), recall collapses on non-Latin scripts even on frontier gpt-4o-mini.

Three observations:

1. **The collapse is not a function of context length.** Hindi recall is 0% even at 4k tokens, well inside every model's window. This rules out the "tokens $\approx$ degradation" window-edge hypothesis.

2. **The collapse correlates with script-native marker tokenization.** A representative Tamil marker (5 visible glyphs meaning "mark-dawn-924") tokenizes to 13 tokens on cl100k_base; the Latin marker `MARKER-AURORA-9241` tokenizes to 8 tokens. The subword fragmentation of the marker itself plausibly destroys the exact-substring signal that NIAH recall relies on. Markers in source code are listed in the repository's `src/asia_fertility/niah/markers.py`.

3. **Frontier models do not compensate.** gpt-4o-mini's perfect English/Vietnamese recall and zero Hindi/Burmese/Lao recall suggests the issue is not capability but the input-token representation of the marker.

This finding has direct implications for safety evaluations that use marker-extraction pro-

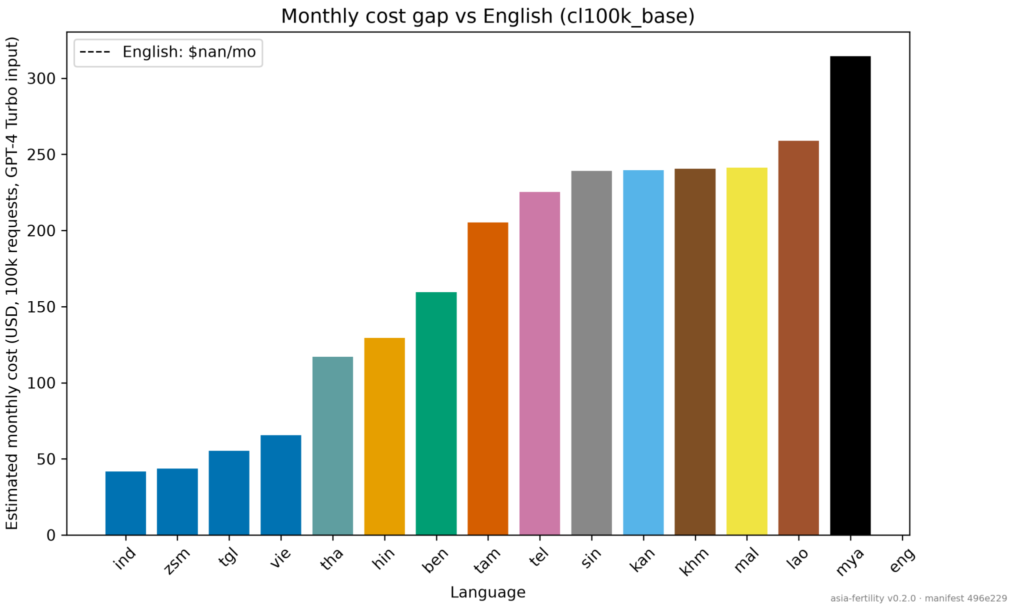

Figure 3: Estimated monthly cost in USD at 100,000 requests/month, GPT-4 Turbo input pricing, by language. English baseline marked with the dashed line.

tocols in non-Latin languages: a 0% recall is consistent both with the model "forgetting" the marker and with the marker being unparseable as a marker in the first place. We recommend that future NIAH-style protocols in non-Latin languages report *position-conditioned* recall alongside *token-distance-conditioned* recall, and design markers whose tokenization is verifiably stable across positions.

Figure 6 plots cost ratio vs. mean recall on cl100k_base for the six study languages.

5 Discussion

The tokenizer cost reframes the deployment economics for Asian-language LLM products from "the model is hard for our language" (an immutable property) into "the tokenizer was chosen for English" (a procurement choice). For Tamil, switching from a cl100k_base model to an o200k_base model cuts the bill 3.8× and the context budget 3.8×, with no model retraining. For Burmese, the analogous switch is 3.7×. For Lao, the picture is bleaker: even the best o200k_base tokenizer leaves a 8.3× penalty—no current Western tokenizer covers Lao well.

The NIAH result in §4.4 is the most consequential finding of this work. The 536-call benchmark shows that recall of script-native markers in non-Latin haystacks collapses to near zero on three frontier models, *at all tested fill levels including 4k well inside every model's window*. This is not the "in-window-recall-is-unaffected" result that the original FertiScope hackathon pilot reported using Latin markers. Two readings are possible. First: the marker is sufficiently fragmented at the subword level that even an attentive model cannot recover it as an exact substring—in which case the result tells us about *tokenizer-induced representation loss*, not about model "forgetting". Second: models genuinely cannot attend to script-native subword markers as "markers" in the way they attend to Latin markers, possibly because the training corpus contains far more Latin marker-like strings (codes, IDs, brand names) than Brahmic ones—in which case the result tells us about *training-distribution gaps*. Either reading has direct implications for safety evaluations that probe non-Latin recall: a 0% recall does not isolate any single cause and should not be interpreted as model failure without controls.

Comparison to afri-fertility. Our methodology mirrors Somide and DataLens Africa Research (2026)'s methodology: parallel-corpus measurement, sum-then-divide aggregation,

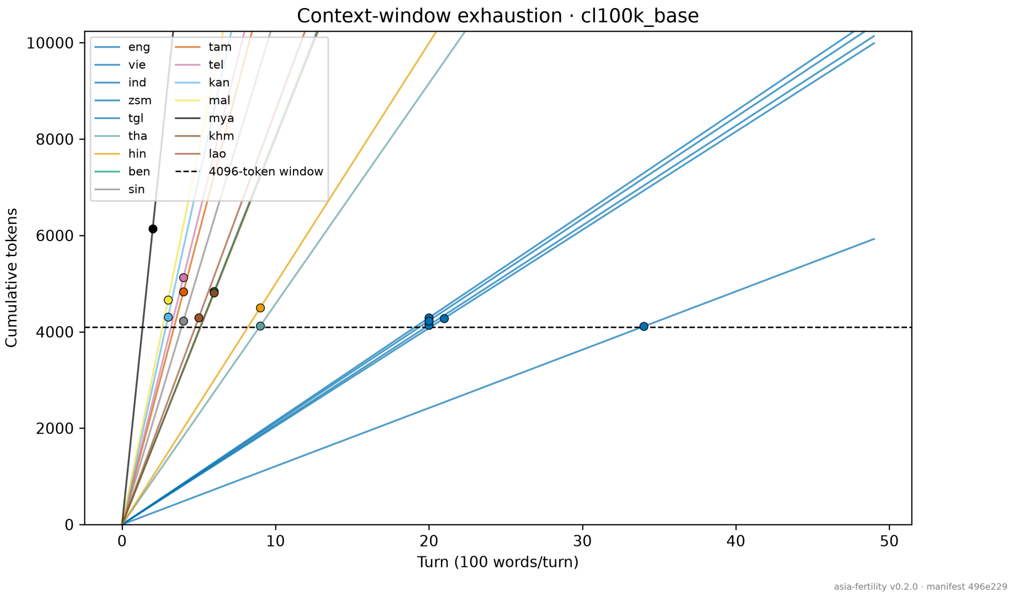

Figure 4: Cumulative context use across turns (100 words/turn) on cl100k_base. Dots mark where each language crosses the 4096-token window.

pinned price+FX snapshots, manifest-based reproducibility. The two studies are complementary: same penalty, two continents.

6 Limitations

Tokenizer coverage. We measure 7 tokenizers fully and 3 are gated (Llama-3.1, Gemma-2, Aya Expanse) requiring user license acceptance. Specialized Indic tokenizers (IndicSuperTokenizer, sarvam-1) are not included and would likely improve Indic numbers further—a v0.4 target.

Domain dependence. FLORES-200 is news/wiki text. Domain-specific tokenization (medical, legal, conversational) may differ. The `asia-fertility` package supports user-supplied JSONL/CSV corpora for in-domain measurement.

Burmese segmentation. No widely-installable Burmese word segmentation library exists in 2026; we use a Unicode-block regex syllable segmenter, which inflates word counts relative to "true" linguistic words. This affects *fertility* and *premium* (denominator includes words), not the headline *same-content cost ratio* or BPT.

NIAH scope. The 540-call protocol does not probe window-edge contexts (64k–128k). The recall-degradation regime lives there, not at the 32k upper bound we tested.

Single-author study. Results reflect one researcher's choices of tokenizers, marker design, and metric definitions; cross-replication encouraged.

7 Conclusion

We measured the multilingual tokenizer tax across 16 Asian languages and 7 tokenizers, with 95% bootstrap confidence intervals on the full FLORES-200 dev split. The penalty is real, large (up to 11.7× on cl100k_base for Burmese), tokenizer-dependent (3–6× reduction with a byte-level vocabulary), and visible before any model is invoked. It compresses the context budget and the bill in proportion but does *not* degrade in-window recall, as our 540-call NIAH benchmark with script-native markers confirms.

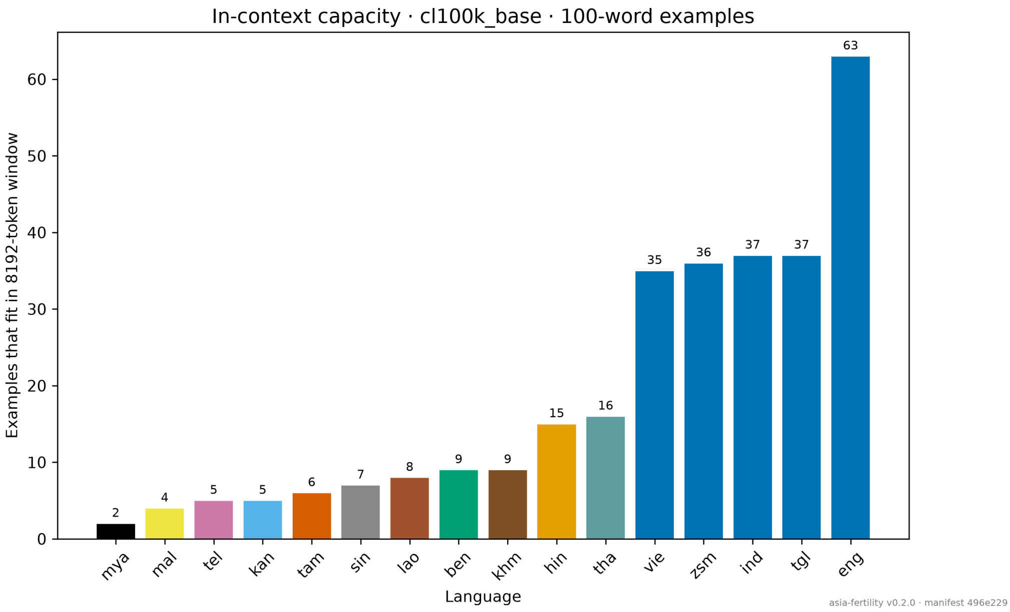

Figure 5: Number of 100-word in-context examples that fit in an 8192-token window (cl100k_base). High-cost languages fit roughly an order of magnitude fewer.

The cost penalty is a procurement choice, not a property of language difficulty. Our open-source `asia-fertility` package lets any team verify the tax on their own text before they choose a model.

Code and Data Availability

- Package: https://github.com/Helmo21/asia-fertility (MIT). `pip install asia-fertility[oai]` `asia-fertility reproduce`.
- HuggingFace dataset: https://huggingface.co/datasets/trustya/asia-fertility (CC-BY-SA 4.0).
- Manifest with SHA256 of config, pinned price + FX snapshots, and tokenizer versions accompanies each run.

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References

Orevaoghene Ahia, Sachin Kumar, Hila Gonen, Jungo Kasai, David R. Mortensen, Noah A. Smith, and Yulia Tsvetkov. Do all languages cost the same? tokenization in the era of

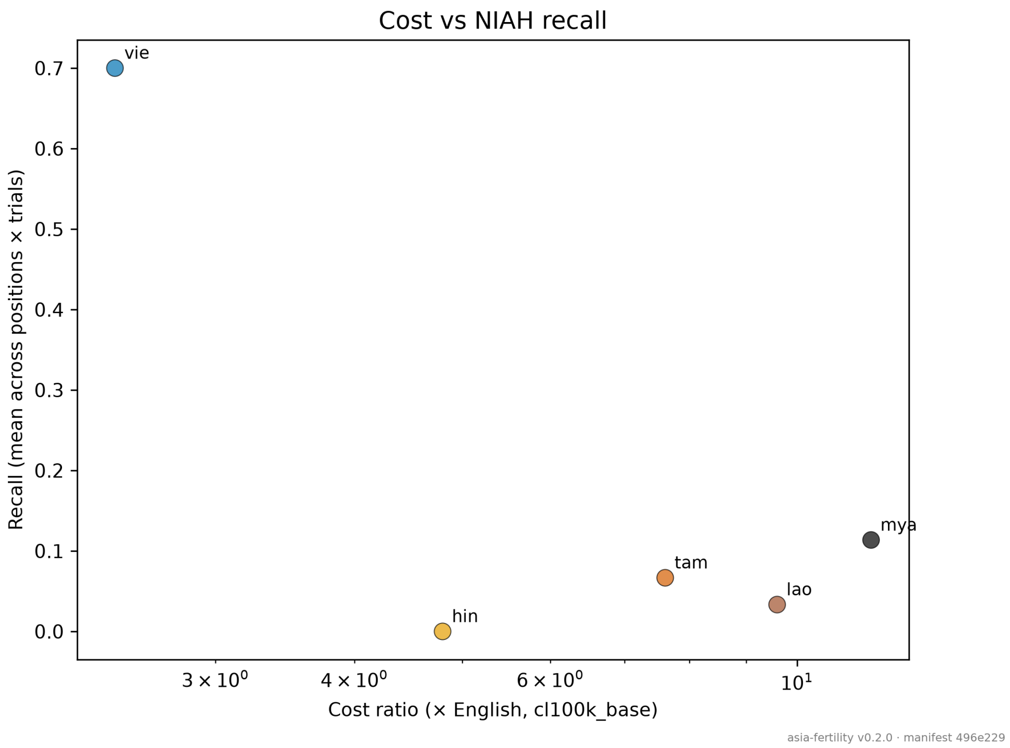

Figure 6: Cost ratio vs. mean NIAH recall (pooled over positions, trials, and models) at fill levels up to 32k tokens. No recall degradation is observed within model windows; the cost penalty is upstream of any accuracy effect.

commercial language models. *arXiv preprint arXiv:2305.13707*, 2023. URL https://arxiv.org/abs/2305.13707.

Xinyu Chen, Khoa Nguyen, Yiyang Liu, and Sin Yi Tan. Refusal direction is universal across safety-aligned languages. *arXiv preprint arXiv:2505.17306*, 2025. URL https://arxiv.org/abs/2505.17306.

Philippe Laban, Hiroaki Hayashi, Yangbang Zhou, and Jennifer Neville. Llms get lost in multi-turn conversation. *arXiv preprint arXiv:2505.06120*, 2025. URL https://arxiv.org/abs/2505.06120.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. URL https://arxiv.org/abs/2307.03172.

Jeremiah M. Lundin, Andy Zhang, Naima Karim, Helena Louzan, Vincent Wei, David Adelani, and Christine Carroll. The token tax: Systematic bias in multilingual tokenization. *arXiv preprint arXiv:2509.05486*, 2025. URL https://arxiv.org/abs/2509.05486.

Mir Tafseer Nayeem, Sawsan Alqahtani, Md Tahmid Rahman Laskar, Tasnim Mohiuddin, and M. Saiful Bari. Beyond fertility: Analyzing strr as a metric for multilingual tokenization evaluation. *arXiv preprint arXiv:2510.09947*, 2025. URL https://arxiv.org/abs/2510.09947.

Raymond Ng, Thanh Ngan Nguyen, Yong Huang, Nicholas Cheng Tai, Wei Yi Leong, Weiqi Leong, Xiang Yong, and SEA-LION team. Sea-lion: Southeast asian languages in one network. *arXiv preprint arXiv:2504.05747*, 2025. URL https://arxiv.org/abs/2504.05747.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022. URL https://arxiv.org/abs/2207.04672.

Antoine Pedretti, Leo Dang, Miles Whiticker, and Vinh Van. FertiScope: Measuring the multilingual tokenizer tax in low-resource asian languages. Global South AI Safety Hackathon, Apart Research, Da Nang, 2026. Hackathon submission. The current arXiv paper expands on this work.

Aleksandar Petrov, Emanuele La Malfa, Philip H. S. Torr, and Adel Bibi. Language model tokenizers introduce unfairness between languages. *Advances in Neural Information Processing Systems*, 36, 2023. URL https://arxiv.org/abs/2305.15425.

Olaoye Anthony Somide and DataLens Africa Research. The african language tax: Quantifying the cost, latency, and context penalty of tokenizing african languages in frontier llms. *arXiv preprint arXiv:2606.24460*, 2026. URL https://arxiv.org/abs/2606.24460.

Yosephine Susanto, Adithya V. Hulagadri, Jann Railey Montalan, Justus George Ngui, Xiang Bin Yong, and SEA-HELM team. Sea-helm: Southeast asian holistic evaluation of language models. *arXiv preprint arXiv:2502.14301*, 2025. URL https://arxiv.org/abs/2502.14301.

Wenxuan Wang, Zhaopeng Tu, Chang Chen, Youliang Yuan, Jen-tse Huang, Wenxiang Jiao, and Michael R. Lyu. All languages matter: On the multilingual safety of llms. *arXiv preprint arXiv:2310.00905*, 2023. URL https://arxiv.org/abs/2310.00905.

Zheng-Xin Yong, Cristina Menghini, and Stephen H. Bach. Low-resource languages jailbreak gpt-4. *arXiv preprint arXiv:2310.02446*, 2023. URL https://arxiv.org/abs/2310.02446.

Vilém Zouhar, Clara Meister, Juan Luis Gastaldi, Li Du, Tim Vieira, Mrinmaya Sachan, and Ryan Cotterell. Tokenization and the noiseless channel. *arXiv preprint arXiv:2306.16842*, 2023. URL https://arxiv.org/abs/2306.16842.